

EUHorizonClient Integration & Reference Guide

Note from Simayy:

Hey! Making EUHorizonClient the single shared source makes total sense. Just a heads-up: a few fields (like budget and eligibility) aren't fully populated yet or act like placeholders because the raw EU API metadata doesn't always return them. In my testing, I also noticed that more complex or strict queries sometimes return empty results. If we decide to adopt it as our shared source, we should test a few test cases together to make sure it covers everything we need.

That being said, if you'd like to wire it in and start testing, here is how you can do it:

1. How to Import and Call It

Imports:

```
from backend.clients.sources.eu_horizon import EUHorizonClient
from backend.schemas.grants import GrantSearchRequest
```

Signature & Call:

```
client = EUHorizonClient()
results = client.search(payload: GrantSearchRequest)
```

GrantSearchRequest Key Parameters:

- **query** (*str* | *None*): Search query string (e.g. 'AI quality inspection').
- **keywords** (*list[str]*): Keyword list if no query is provided (e.g. ['ai', 'robotics']).
- **limit** (*int*, *default=3*): Max number of grant results to return (1 to 25).
- **only_open** (*bool*, *default=False*): Filters for open calls (verifies status and deadline ≥ today).
- **action_type** (*str* | *None*): Case-insensitive action type filter (e.g. 'RIA', 'IA').
- **budget_min** / **budget_max** (*int* | *None*): Budget bounds in EUR (e.g. budget_min=50000).

2. What It Returns (Exact Fields per Grant)

Returns a list[GrantResult]. Each result object contains:

Field	Type	Description	Example
id	str	Topic / Call identifier	HORIZON-CL2-2026-EDU-01
title	str	Official call topic title	Digital education innovation...
source	str	Source identifier system	eu_horizon
summary	str None	Text summary (cleaned HTML, max 600 chars)	Supports edtech initiatives...
amount	str None	Formatted budget overview range	EUR 500 000 - EUR 1 000 000
deadline	str None	Application deadline (YYYY-MM-DD)	2026-11-15
match_explanation	str None	Key match details summary	Matched Horizon topic...; RIA; deadline 2026-11-15
url	str None	Direct topic URL on EU portal	https://ec.europa.eu/...

3. Setup, API Keys & Dependencies

- **API Keys / Auth: None required.** Uses the public EU SEDIA Search API (<https://api.tech.ec.europa.eu/search-api/prod/rest/search>).
- **Dependencies:** Requires httpx and pydantic (already in project requirements).
- **Testing Parameter:** EUHorizonClient(transport=...) accepts an optional custom httpx.BaseTransport for mocking HTTP in unit tests.

4. Quick Usage Example

```
from backend.clients.sources.eu_horizon import EUHorizonClient
from backend.schemas.grants import GrantSearchRequest

client = EUHorizonClient()

request = GrantSearchRequest(
    query="robotics AI",
    limit=5,
    only_open=True,
    action_type="RIA",
)

grants = client.search(request)

for g in grants:
    print(f"[{g.id}] {g.title} | Budget: {g.amount} | Deadline: {g.deadline}")
```